

Coding-Shuttle Java Development

Module 1 - Introduction To Spring & Spring Boot

1. Introduction to Spring Framework
2. Setting Up a Spring boot Project
3. Beans in Spring Framework
4. Dependency Injection in Spring Framework
5. Spring Boot vs Spring Framework, A Comparison
6. Auto Configuration, Application Context and Internal Working of a Spring Boot Application
7. Maven Build Tool

Module 2 - Spring Boot MVC and RESTful APIs

1. Introduction to Spring Boot , MVC Architecture , Tomcat and Dispatcher Servlet
2. The Presentation Layer , DTO & Controllers
3. The Persistence Layer and JPA Repository
4. The Service Layer ,Writing our business logic
5. PUT,PATCH and DELETE Mappings in Spring Web MVC
6. Input Validation Annotations, Creating Custom Annotation for Validation
7. Exception Handling in Spring Boot Web MVC
8. Transforming API response in Spring Web MVC
9. Extra know- JAVA 8 Lambda And Stream API

Module 3 - Hibernate and Spring Boot Data JPA

1. Installing MySQL for Database and DBeaver for Analytics

2. Hibernate ORM,JPA , Entities and Tables in Spring Data JPA
3. Spring Data JPA Interfaces and Dynamic Query Methods.
4. Sorting and Pagination in Spring Data JPA.
5. Projection in Spring Data JPA.
6. Hibernate Entity Lifecycle, Entity Manager and Persistence Context
7. OneToOne, OneToMany, ManyToOne, ManyToMany Mappings in Spring Data JPA
8. Cascading Types and Transactional in Relational Queries.
9. FetchType and Orphan Removal in Relational Queries, N+1 Query Optimization.

Module 4 - Production ready Spring Boot Features

1. Setting up a Spring Boot Project with Spring Boot Web MVC and Spring Data JPA
2. Spring Boot Dev tools
3. Spring Boot Basic Auditing and Advanced Auditing with Hibernate Envers.
4. Making third party API call in Rest Client
5. Logging in Spring Boot Applications.
6. Spring Boot Actuator.
7. OpenAPI and Swagger to Generate the API Documentation.

Module 5 - Spring Security Fundamentals

1. Security Attacks like CSRF,XSS,SQL Injection
2. Internal Working of Spring Security.
3. Spring Security Web Config- SecurityFilterChain Configuration.
4. Understanding JWT ,Creation and Verification of JWT tokens.
5. Signup and Login using JWT
6. Authentication Requests using JWT.
7. Spring Security Exception Handling.

Module 6 - Spring Security Advanced

1. JWT-Refresh Token & Access Token based Authentication
2. Google OAuth2 Client Authentication in Spring Security.
3. User Session Management with JWT.
4. Role based Authorization in Spring Security.
5. Granular Authorization with Authorities and Permissions in Spring Boot Security.
6. Security Method Annotations- @Secured and @PreAuthorized.

Module 7 - Spring Boot Testing

1. Introduction to Testing in Spring Boot.
2. Understanding JUnit Testing Framework and AssertJ Library
3. Unit Testing Vs Integration Testing.
4. Unit Testing the Persistence Layer and Setting up TestContainer.
5. Understanding Mocking with the Mockito Library.
6. Using Testing the Service layer using Mockito.
7. Integration Testing in Spring Testing.
8. JaCoCo Test Report Generation.

Module 8 - Spring Boot Deployment with CI/CD

1. Setup Production Database with RDS in AWS.
2. Spring Profile- Creating Dev,prod environments in Spring Boot.
3. Deploy on Elastic BeanStalk with Load Balancer, EC2, AutoScaling and RDS Connection.
4. Setup CI/CD with AWS CodePipeLine and CodeBuild with Github hooks.
5. Database Migration in Production using Flyway.

Module 9 - Spring AI, RAG and Tool Calling

1. The Mechanics of Generative AI.

2. Introduction to Spring AI and ChatClient.
3. Embedding and Vector Search.
4. RAG (Retrieval Augmented Generation).
5. Mastering Advisors- The Interceptors of Spring AI.
6. Tool Calling in Spring AI.

Module 10 - Aspect-Oriented Programming

1. Introduction to Aspect-Oriented Programming.
2. Declaring Pointcut in AOP.
3. Declaring Advice in AOP.
4. Spring Proxy and Internal working of AOP.
5. Real World use cases of AOP.

Module 11 - Caching and Concurrent Transaction Management

1. Introduction to Caching.
2. Spring Boot default Caching.
3. Redis Cache in Spring Boot.
4. Database Transactions and ACID Properties in RDBMS.
5. Concurrent Transaction management and Transaction isolation levels.
6. Transactional Annotation in Spring Boot.
7. Optimistic and Pessimistic transaction locks in Spring Boot Data JPA

Module 12 - Introduction To Microservice Architecture

1. Introduction to Microservice Architecture.
2. Setting up the Inventory Management System.
3. Service Registration and Service Discovery with Eureka.
4. Spring Cloud API Gateway.
5. Open Feign Microservice communication.
6. Circuit Breaker ,Retry and Rate Limiter with Resilience4J

Module 13 - Advanced Microservice Concepts

1. Introduction to API Gateway Filter.
2. Authentication in the API Gateway using custom GatewayFilters.
3. Centralized Configuration Server using Github.
4. Refresh Configuration without Restart using Centralized Configuration.
5. Distributed Tracing using Zipkin & Micrometer.
6. Centralized Logging with the ELK Stack.

Module 14 - Apache Kafka in Spring Boot

1. Introduction to Kafka.
2. Apache Kafka Architecture (Internal Working of Kafka).
3. Installing Kafka and Kafka Visualisation tool
4. Configuration Kafka With Spring Boot.
5. Advanced Configuration Kafka With Spring Boot.
6. Kafka Schema Registry with Confluent.

Module 15 - Docker with Spring Boot

1. Introduction to Docker
2. Docker Installation and Docker commands.
3. Building Docker images using Docker files and Maven plugin.
4. Managing Multiple docker container using Docker Compose.
5. Dockerize the Microservices Architecture with Docker compose.

Module 16 - Kubernetes Components

1. Introduction to Kubernetes.
2. Installing Kubernetes Cluster using Minikube.
3. Pods, Namespace, Deployment, ReplicaSet
4. Services in Kubernetes.
5. Persistent Volumes and Persistent Volume Claim.

6. Stateful State in Kubernetes.

Module 17 - Kubernetes Advanced

1. Google Kubernetes Engine Setup with gCloud CLI.
2. Database Setup with GKE StatefulSet.
3. Microservice setup with Kubernetes DNS.
4. Ingress Endpoints in Kubernetes.
5. ConfigMap and Secrets in Kubernetes.
6. Kubernetes CI/CD with Github Actions.

Module 18 - Async and Blocking Task Scheduling in Spring Boot

1. Java Executor Framework.
2. Task Scheduling in Spring Boot.
3. Asynchronous Task Scheduling in Spring Boot.
4. Tomcat Server Threading Model.